

Collage of Computer Science and Engineering Department of Software Engineering



Internet of Things (IoT) CCSW-432

Team members:

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Dr. Nadia

Project Overview

MediGlow is a smart IoT-based medication reminder system designed to help elderly people and adults manage their daily medications and vitamin intake efficiently. Forgetting to take medication on time is a common problem that can lead to serious health complications, especially among elderly individuals.

The MediGlow system provides timely visual and sound alerts to remind users when it is time to take their medication. In addition, the system monitors medication levels and expiration dates to ensure safe and consistent usage. Medication data is stored and monitored through cloud services, allowing improved tracking and awareness.

2. Problem Statement

Many people, particularly elderly individuals, face difficulties in managing their medication schedules due to memory issues, complex medication routines, or lack of technological skills. Missing doses, taking expired medication, or running out of medication without notice can lead to serious health risks.

While smartphone applications exist, they often require technical knowledge and constant interaction, making them unsuitable for elderly users. Therefore, there is a need for a simple, reliable, and elderly-friendly medication reminder system that does not depend on screens or complicated interfaces.

3. Project Objectives

The primary objective of the MediGlow Smart Medication Reminder System is to improve medication adherence and safety for elderly individuals and adults by providing a simple, reliable, and intelligent reminder solution. The system is designed to address common challenges related to medication management without relying on complex technology or user interfaces.

The specific objectives of this project are as follows:

1. Improve Medication Adherence

To ensure users take their medications and vitamins at the correct time every day by providing clear visual and audio reminders. This helps reduce missed doses and irregular medication intake, which can negatively affect health outcomes.

2. Support Elderly Users with a User-Friendly Design

To design a medication reminder system that is easy to use for elderly people by avoiding complicated screens, mobile applications, or difficult interactions. The system relies on simple LEDs and a buzzer, making it suitable for users with limited technical experience.

3. Reduce Medication Errors

To minimize common medication-related errors such as forgetting doses, taking medication late, or consuming expired medicine. By alerting users about medication schedules, expiration dates, and low medication levels, the system promotes safer medication practices.

4. Enhance Medication Safety

To improve overall medication safety by continuously monitoring medication availability and expiration status. The system helps prevent health risks caused by taking expired or insufficient medication.

5. Assist Caregivers and Family Members

To provide caregivers and family members with peace of mind by enabling cloud-based monitoring of medication status. This allows them to track whether medications are taken properly and intervene when necessary.

6. Promote Independent Living

To support elderly individuals in maintaining independence by reducing their reliance on constant supervision for medication management. MediGlow enables users to manage their medications confidently and independently.

7. Utilize Smart IoT Technology Effectively

To demonstrate the practical application of Internet of Things (IoT) technology in healthcare by integrating sensors, microcontrollers, and cloud platforms into a real-world medical solution.

8. Provide an Affordable Healthcare Solution

To develop a cost-effective medication reminder system using readily available hardware components, making it accessible to a wide range of users without high financial burden.

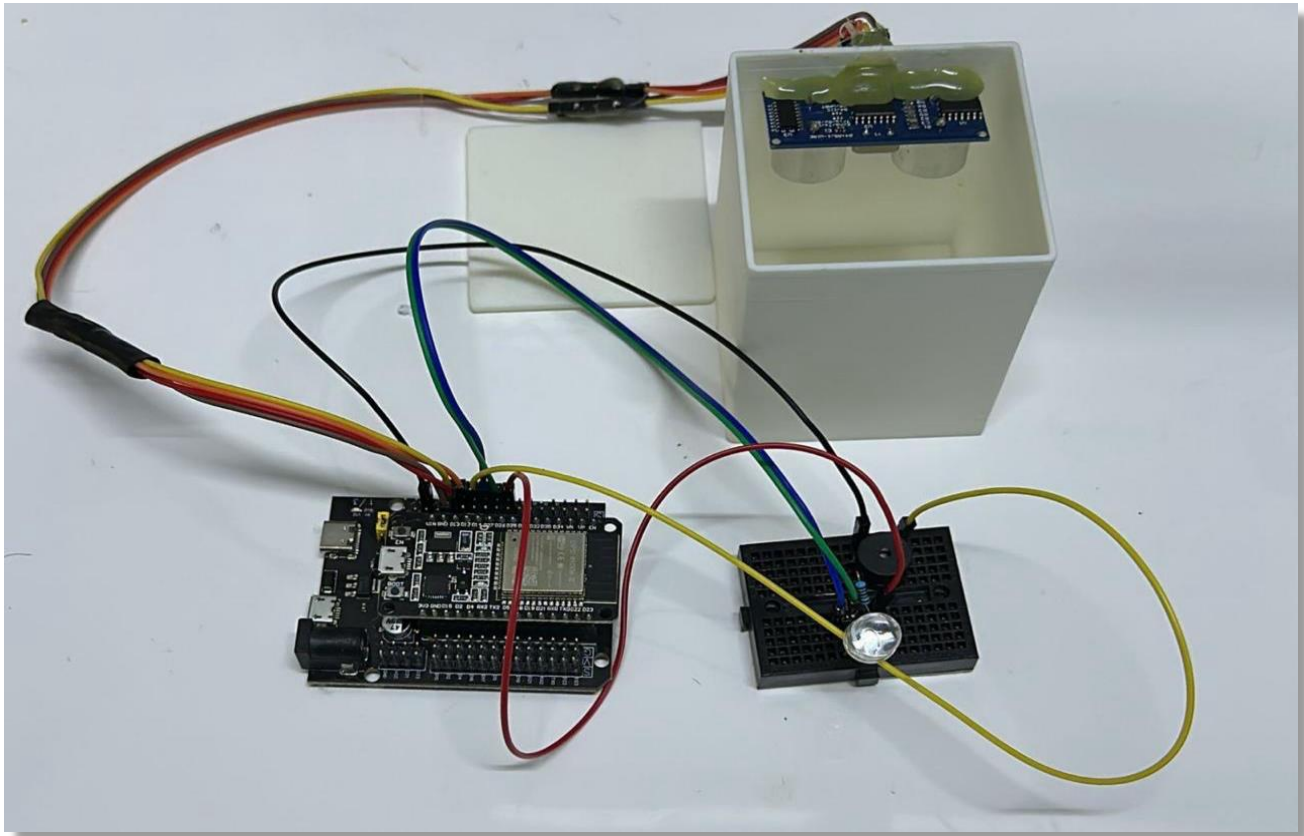
Hardware component:



- Buzzer
- RGB LED
- Resistor
- Pin Headers
- ESP32 Microcontroller
- Ultrasonic Sensor
- Breadboard
- Jumper Wires

Hardware connection:





Software component:



IFTTT cloud



ThingSpeak



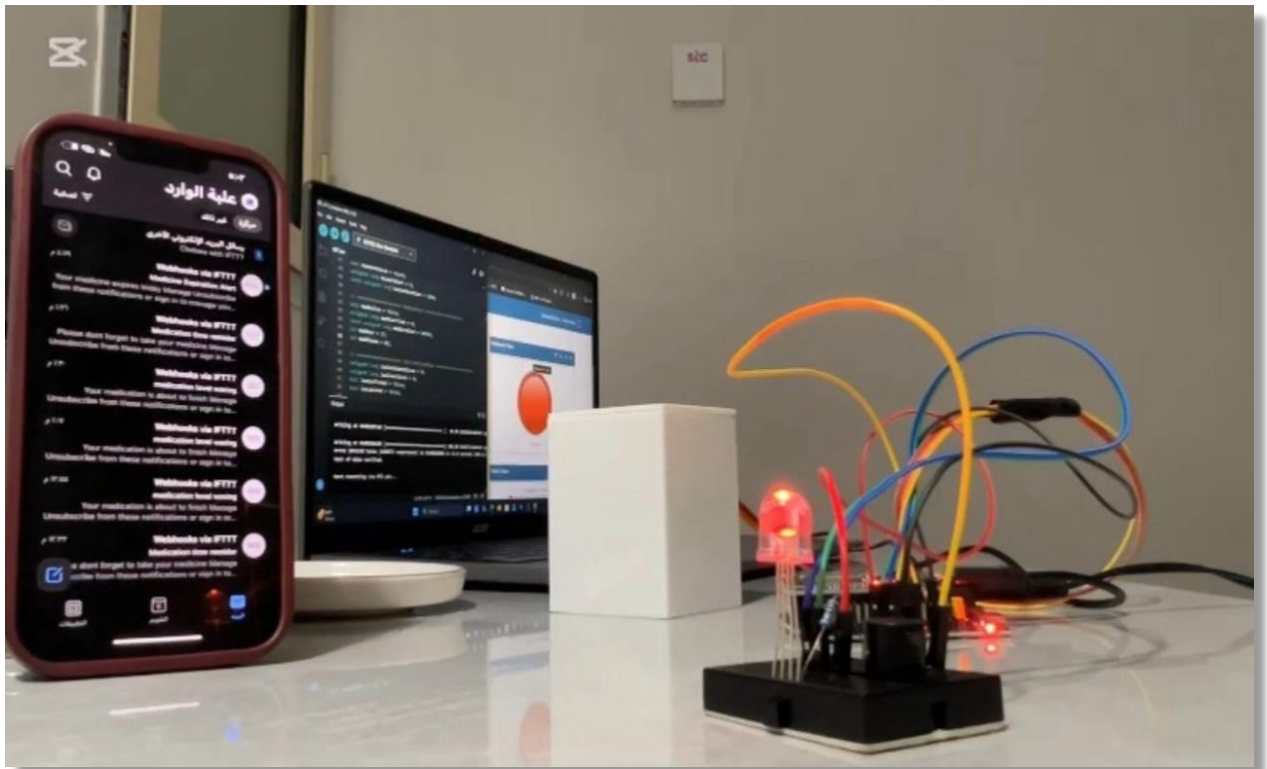
Arduino IDE

Video:



-A video showing how the device works

Test:



-Medication time



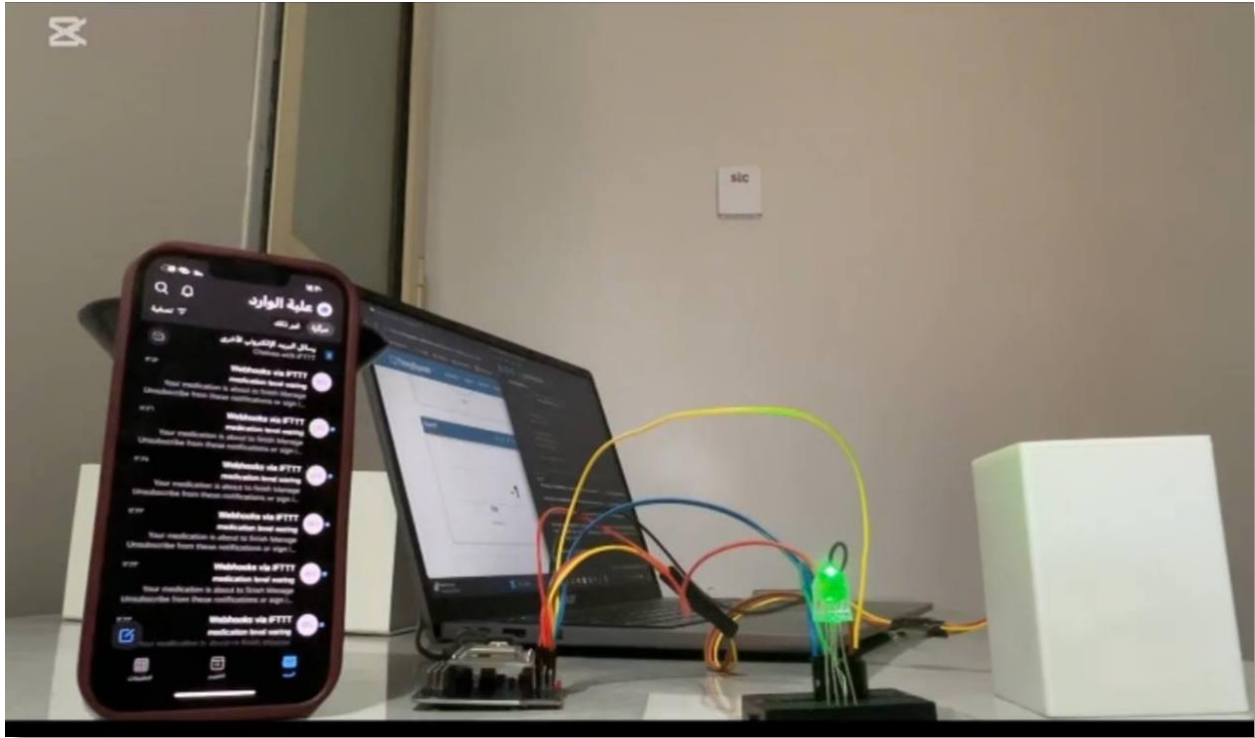
-The normal situation in this picture



-in this picture, a message appeared telling you to take the medicine.



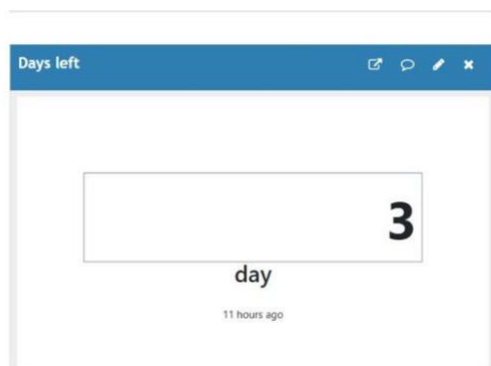
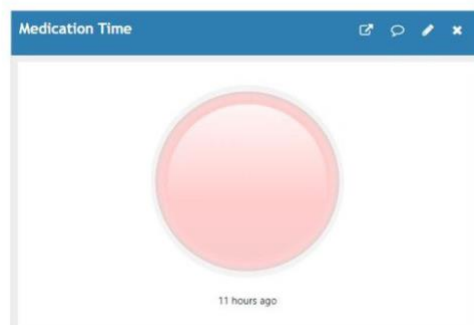
-This picture shows the situation when the medicine is nearly finished



-This image shows that the medication is expired



Last entry: about 11 hours ago
Entries: 1450



Snipping Tool
Screenshot copied to clipboard

Channel 3198844

Channel ID: **3198844**

Author: mwa0000039881737

Access: Public

[Private View](#)

[Public View](#)

[Channel Settings](#)

[Sharing](#)

[API Keys](#)

Write API Key

Key

8YVCQ7URU0KT6IQS

[Generate New Write API Key](#)

[Read API Keys](#)

Apps - ThingHTTP - ThingSpeak

Inbox - lot Project - Outlook

medication_time - ifTTT

maker.iftt.com/trigger/medic...

https://thingspeak.mathworks.com/apps/thinghttp/229631/edit

https://chibetales...

Gotham S04E22 WE...

Qualcomm Careers...

What is the differen...

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50pcs 1K 10K 100K...

All Bookmarks

ThingSpeak™

Channels Apps Devices Support

Commercial Use How to Buy MS

Apps / ThingHTTP / medication_time / Edit

Name

medication_time

API Key

WZF0FRFB9VM4VL3H

URL

https://maker.iftt.com/trigger/medication_time/with/key:bGm

HTTP Auth Username

HTTP Auth Password

Method

GET

Content Type

HTTP Version

1.1

Host

Name

Value

remove header

Help

ThingHTTP Settings

- Name:** Enter a unique name for your ThingHTTP request.
- API Key:** Auto generated API key for the ThingHTTP request.
- URL:** Enter the address of the website you are requesting data from or writing data to starting with either `http://` or `https://`.
- HTTP Auth Username:** If your URL requires authentication, enter the username for authentication to access private channels or websites.
- HTTP Auth Password:** If your URL requires authentication, enter the password for authentication to access private channels or websites.
- Method:** Select the HTTP method required to access the URL.
- Content Type:** Enter the MIME or form type of the request content. For example, `application/x-www-form-urlencoded`.
- HTTP Version:** Select the version of HTTP on your server.
- Host:** If your ThingHTTP request requires a host address, enter the domain name here. For example, `api.thingspeak.com`.
- Headers:** If your ThingHTTP request requires custom headers, enter the information here. You must specify the name of the header and a value.
- Body:** Enter the message you want to include in your request.
- Parse String:** If you want to parse the response, enter the exact string to look for in the response data.

Example

Send an HTTP GET request and parse the response for an element corresponding to the current bitcoin price

1. Enter `https://coinmarketcap.com/currencies/bitcoin/` in the **URL** field
2. Enter `GET` in the **Method** field
3. In the **Parse String** field, enter `string(/html/body/div[1]/div[2]/div/div[3]/div/div/div[1]/div/section/div/div[2]/span)`
4. Save and attach the HTTP request to a reaction you set up in the React or TimeControl apps.

Learn More



Apps - React - ThingSpeak IoT

Inbox - IoT Project - Outlook

medication, time - IFTTT

maker.ifttt.com/trigger/LOW_L...

https://thingspeak.mathworks.com/apps/reacts

https://ch3.be.taleo...Gotham S04E22 WE...Qualcomm Careers...What is the differen...920023959reference designer.c...Nile.com.my: Malay...50pcs 1K 10K 100K...All Bookmarks

ThingSpeak™ChannelsAppsDevicesSupportCommercial UseHow to BuyMS

Apps / React

New React

Name	Created	Last Ran
☒ medication time ViewEdit	2025-12-10	2025-12-11 1:03 pm
☒ level ViewEdit	2025-12-10	2025-12-12 8:38 pm
☒ expiration ViewEdit	2025-12-10	2025-12-11 1:04 pm

Help

React works with [ThingHTTP](#) and [MATLAB Analysis](#) apps to perform actions when channel data meets a certain condition. For example, you can have a mobile app report your latitude and longitude to a ThingSpeak channel. When your position is within a certain distance of your house, have ThingHTTP turn on your living room lights.

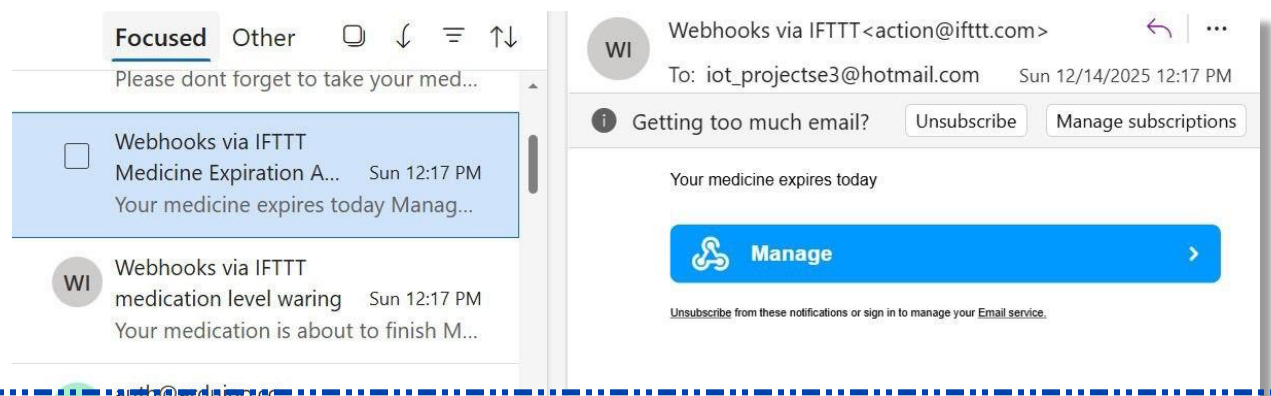
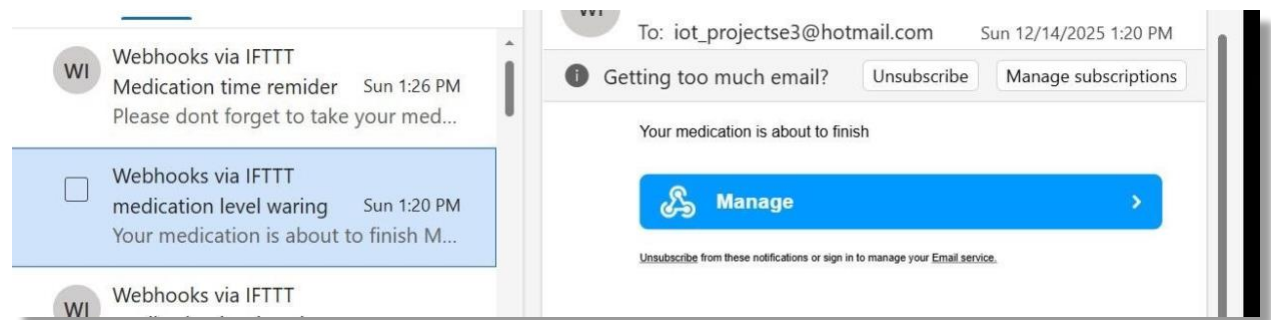
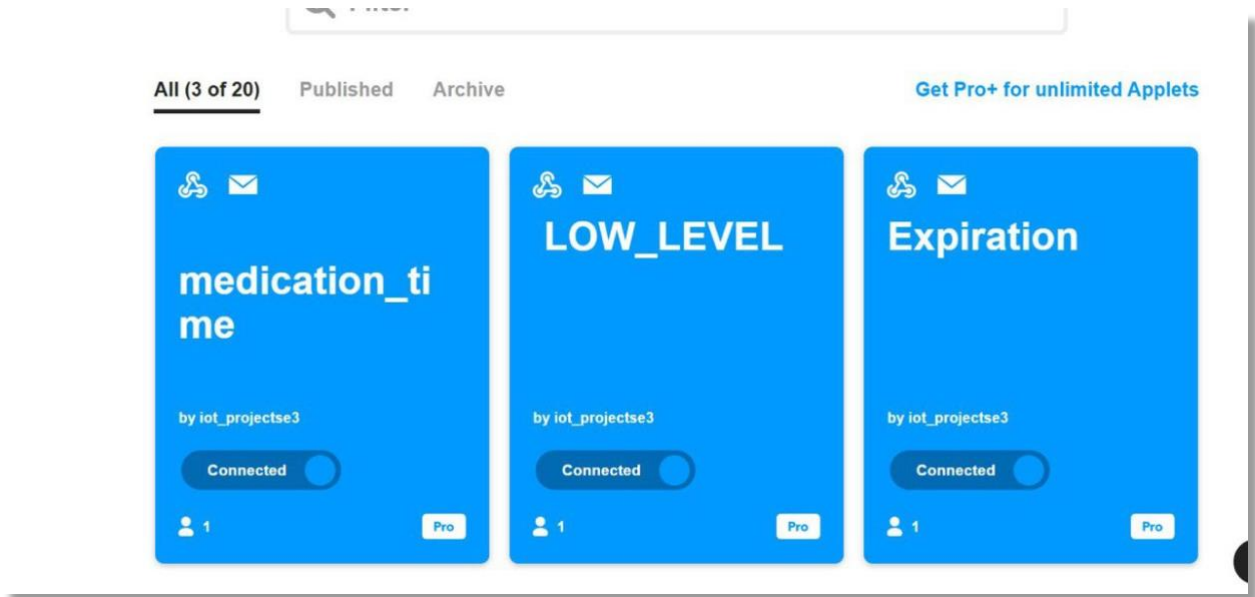
To create a new reaction, click [New React](#).

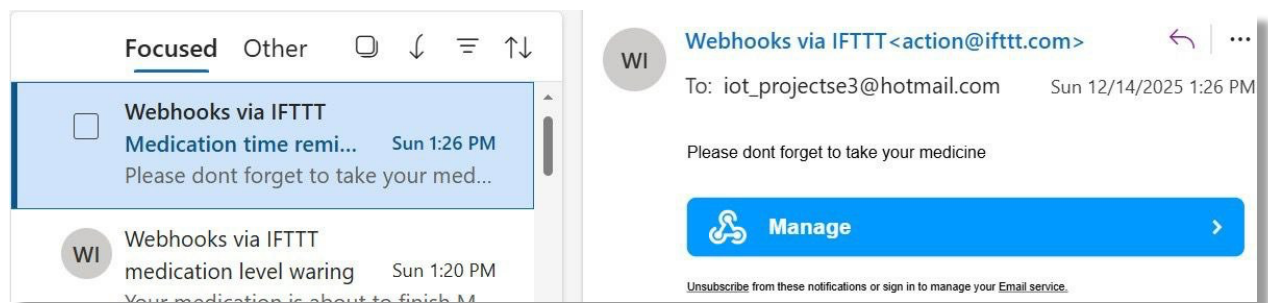
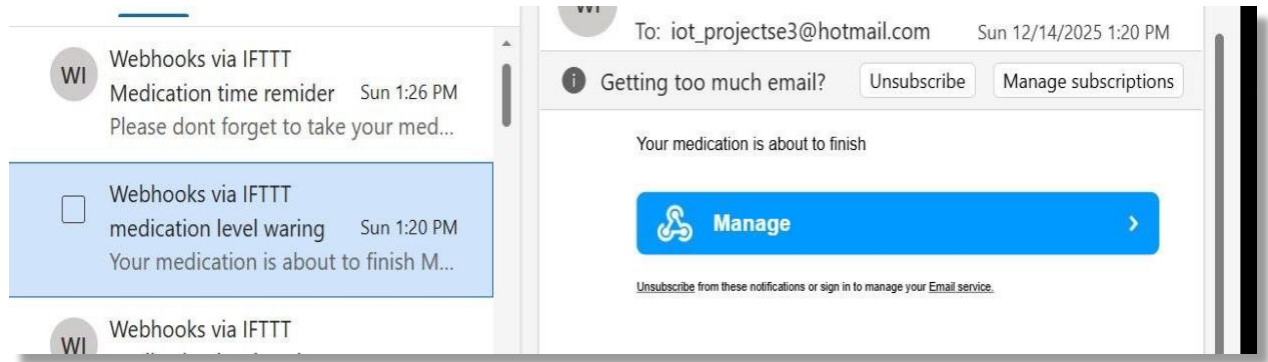
[Learn more](#)

Examples

[Act On Your Data](#)



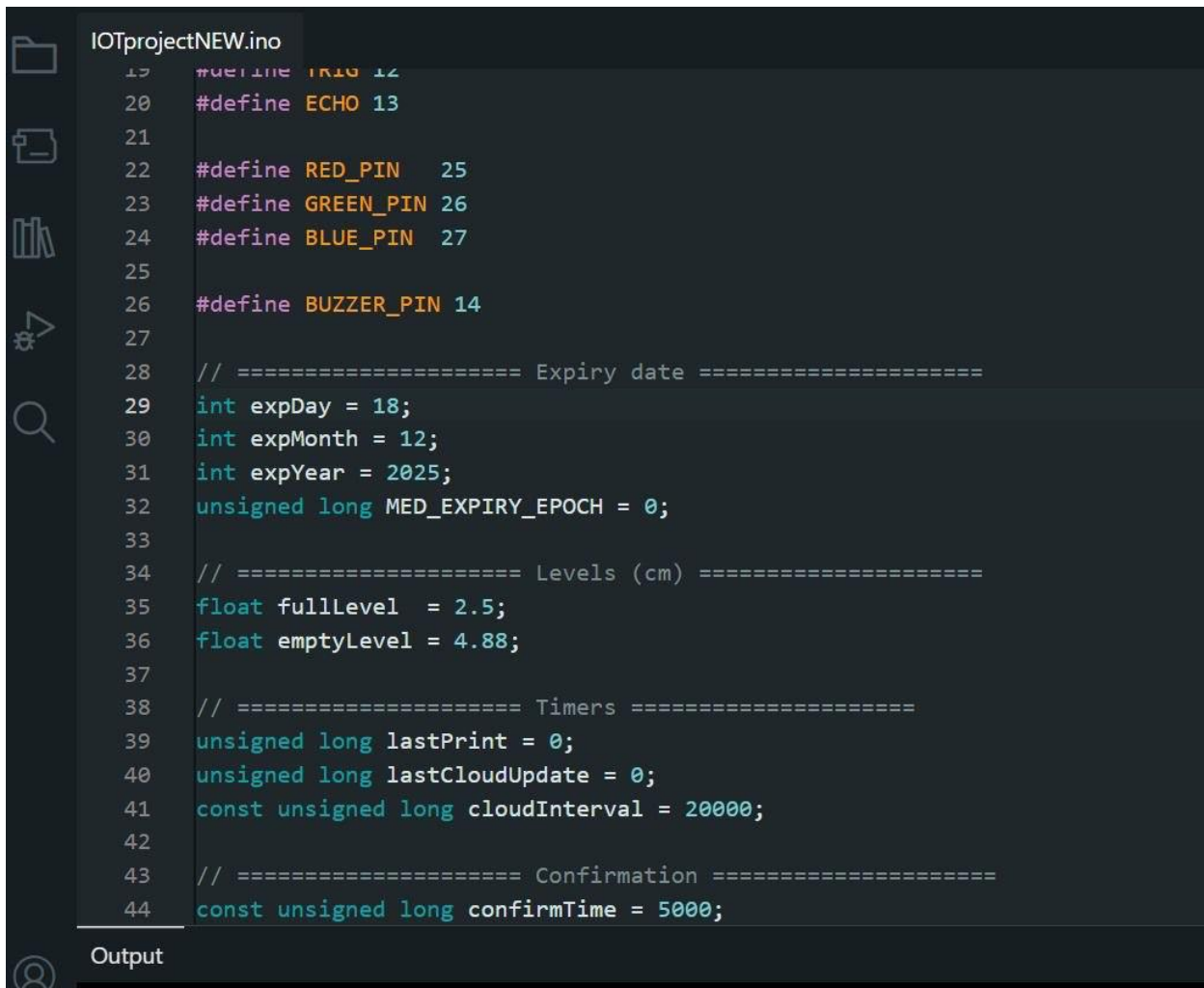




Code:

```
IOTprojectNEW.ino
1  #include <WiFi.h>
2  #include <WiFiUdp.h>
3  #include <HTTPClient.h>
4  #include <time.h>
5
6  // ===== WiFi =====
7  const char* ssid = "Med";
8  const char* password = "11223366m";
9
10 // ===== ThingSpeak =====
11 String thingspeakApiKey = "8YVCQ7URU0KT6IQS";
12
13 // ===== Time (GMT+3) =====
14 const char* ntpServer = "pool.ntp.org";
15 const long  gmtOffset_sec = 3 * 3600;
16 const int   daylightOffset_sec = 0;
17
18 // ===== Pins =====
19 #define TRIG 12
20 #define ECHO 13
21
22 #define RED_PIN 25
23 #define GREEN_PIN 26
24 #define BLUE_PIN 27
25
26 #define BUZZER_PIN 14
```

Output



```
IOTprojectNEW.ino
19 #define TRIG 12
20 #define ECHO 13
21
22 #define RED_PIN 25
23 #define GREEN_PIN 26
24 #define BLUE_PIN 27
25
26 #define BUZZER_PIN 14
27
28 // ===== Expiry date =====
29 int expDay = 18;
30 int expMonth = 12;
31 int expYear = 2025;
32 unsigned long MED_EXPIRY_EPOCH = 0;
33
34 // ===== Levels (cm) =====
35 float fullLevel = 2.5;
36 float emptyLevel = 4.88;
37
38 // ===== Timers =====
39 unsigned long lastPrint = 0;
40 unsigned long lastCloudUpdate = 0;
41 const unsigned long cloudInterval = 20000;
42
43 // ===== Confirmation =====
44 const unsigned long confirmTime = 5000;
```

Output